INTERNNOVA | WEEK 6 ASSIGNMENT

Zomato Restaurant Analytics
End-to-End Data Analytics Project

GitHub documentation, data preparation, exploratory analysis, Power BI dashboard and business recommendations

Project	Zomato Restaurant Analytics Dashboard
Tools	Power BI, Power Query, DAX, Excel, GitHub
Repository	github.com/vinaygadde2020/Zomato-Restaurant-Analytics-Dashboard-Power-BI-Project
Source data used	Cleaned Zomato dataset; 6,801 rows × 11 columns
Report scope	Week 6 end-to-end project documentation based on the existing Week 5 project

Prepared using evidence from the supplied Week 5 report and the repository screenshot. Items not verified are explicitly identified.

1. Executive Summary

This project analyzes restaurant listings in Bangalore using a cleaned Zomato dataset and a Power BI dashboard. The dashboard summarizes average rating, distinct restaurant names, locations and approximate cost for two people, and includes visuals for votes, rating categories, cuisines, location-by-rating counts and table-booking availability. The existing GitHub repository contains the Power BI file, cleaned Excel dataset, dashboard screenshot and README.

The analysis indicates that Good and Very Good are the most common rating categories, North Indian is the most frequent cuisine entry, and only 19.51% of cleaned listing rows indicate table-booking availability. These are descriptive associations in this dataset and should not be interpreted as proof of cause and effect.

2. Problem Statement

Restaurant customers and business stakeholders need a concise way to explore restaurant ratings, cuisines, customer votes, location coverage, approximate cost and table-booking availability. The project uses data preparation, exploratory analysis and interactive visualizations to summarize these characteristics and identify areas for further investigation.

3. Dataset Description

The cleaned dataset contains 6,801 rows and 11 columns in the supplied workbook. The Power BI model uses eight imported columns in the central Zomato_Fact table: restaurant name, online-order flag, table-booking flag, rating text, votes, location, cuisines and approximate cost for two people. Three calculated columns provide numeric rating, rating category and full location. The original raw zomato.csv file was not supplied with the reviewed materials, so raw row counts and raw missing-value totals cannot be reported.

4. Data Cleaning and Preparation

The Power Query steps recorded in the PBIX include promoting headers, setting data types, removing errors, removing blank rows, removing duplicate URLs, dropping unneeded columns, filtering incomplete records and retaining eight analysis columns. Incomplete rows were filtered out rather than imputed. The cleaned workbook has no missing values across its 11 columns, according to the Week 5 report.

A quality concern remains: 2,258 rows (33.20%) are exact duplicates across the eight retained columns. These may represent repeated listings or separate outlets that share the same retained values; the available files cannot establish which. Keep a unique listing identifier such as URL in future cleaning steps until duplicate handling has been validated.

Another limitation is that all 6,801 retained rows have online_order = Yes. Therefore, this cleaned dataset cannot support a useful comparison between online and offline ordering.

5. Exploratory Data Analysis

Measure	Verified result
Cleaned listing rows	6,801
Distinct restaurant names	1,658
Distinct locations	80
Average numeric rating	3.90
Average approximate cost for two	629.09 (currency not specified in source)
Total votes	3,661,283
Rows with table booking = Yes	1,327 / 6,801 = 19.51%

Measure	Verified result
Very Good rating category	3,153 rows = 46.36%; largest category, not a majority
Good rating category	3,093 rows = 45.48%

Additional findings from the supplied analysis: Onesta has the highest total votes among restaurant names at 136,666; North Indian appears in 2,733 listing rows when cuisine combinations are split; and the above-1,000 cost band defined for the report has the highest average votes per listing. Cost-band thresholds are analysis choices, not original dataset fields.

6. Visualizations and Dashboard

The existing dashboard includes four KPI cards; a bar chart of the top ten restaurant names by votes; a donut chart for rating-category distribution; a treemap of cuisine entries; a location-by-rating matrix; a table-booking donut; and slicers for location, rating category and cuisine. The screenshot shows all three slicers set to All. The interactive behavior was not tested in the supplied report, so slicer and cross-filter operation must be tested in Power BI Desktop before claiming it has been verified.

Suggested chart explanations for the report: (1) Rating-category donut — Good and Very Good dominate the retained listings. (2) Cuisine treemap — North Indian is the most frequent cuisine entry. (3) Votes bar chart — Onesta leads by aggregated votes across its listings. (4) Booking donut — 19.51% of listing rows indicate table booking. (5) Location-rating matrix — rating-category counts vary by neighborhood; the matrix uses distinct restaurant names while several other visuals count listing rows.

7. Business Insights

1. Very Good is the largest rating category at 46.36%, but it is not a majority; Good is close behind at 45.48%.

2. North Indian appears in 40.19% of listing rows when cuisine combinations are split; this suggests a crowded cuisine segment in this dataset.

3. Desserts + Beverages has an average rating of 4.15 across 110 rows, while North Indian + Chinese averages 3.59 across 179 rows.

4. The top ten restaurant names account for about 17.0% of all votes; Onesta has 136,666 votes across 32 rows.

5. Only 19.51% of listing rows offer table booking. Rows with booking have higher average ratings and votes in this dataset, but the association does not prove booking causes better engagement.

6. The five largest locations by listing rows—BTM, Koramangala 5th Block, Indiranagar, JP Nagar and Jayanagar—together account for 33.1% of rows.

8. Recommendations

1. Pilot table-booking onboarding. Start with well-rated restaurants in areas where booking availability is low, then track changes in listing engagement. Treat this as a pilot, not a guaranteed causal improvement.

2. Differentiate cuisine offerings. Promote discovery of well-rated dessert and beverage listings and help crowded cuisine segments investigate service consistency, menu differentiation and customer feedback.

3. Improve data reliability. Preserve a unique listing identifier during cleaning, document whether visuals count rows or distinct restaurant names, define cost bands explicitly and correct the rating relationship in the model.

9. GitHub and Project Documentation

The repository screenshot shows a public GitHub repository with README.md, Dashboard Screenshot.png, zomato.pbix and zomato_cleaned.xlsx. The commit history screenshot shows three commits: Initial commit, Add files via upload and Update README.md. This verifies that the repository has three commits, but the history alone does not prove that the Git workflow operations clone, add, commit, push and pull were all practiced. Complete and document any remaining workflow practice before submission.

Recommended repository structure (adapt paths to files that actually exist):

```
README.md
Dashboard Screenshot.png
zomato.pbix
zomato_cleaned.xlsx
report/InternNova_Week6_Final_Project_Report.pdf
presentation/InternNova_Week6_Project_Presentation.pptx
```

10. Limitations and Items to Verify

• The original raw dataset was not available in the reviewed materials; raw cleaning counts cannot be stated.
• Exact duplicates on retained columns need investigation before deduplication decisions are finalized.
• All retained online_order values are Yes, so online/offline comparison is not possible.
• The Week 5 report flags a likely incorrect relationship between Zomato_Fact[rate_numeric] and Rating_Table[Rating Category]. Recreate the relationship using the matching Rating Category columns and test it.
• The existing Highest Rated Restaurant measure uses the text rating field and needs correction/testing if retained.
• Slicer and cross-filter interactions were not tested in the source report.
• The currency of approximate cost is not specified; do not add a currency symbol without confirming it.

11. Conclusion

This project demonstrates an end-to-end analytics workflow using a cleaned restaurant dataset, Power Query transformations, DAX calculations, a Power BI data model and dashboard visuals. The dashboard summarizes rating, location, cuisine, votes and table-booking patterns. The next priorities are to document the repository clearly, validate the model relationship and slicers, and include the final report and a short presentation in the GitHub submission.

Evidence note: Results and model observations above are derived from the uploaded 16-page Week 5 report. Repository contents are based on the screenshot supplied in this conversation. This report does not claim that untested Power BI interactions or Git workflow operations have been completed.